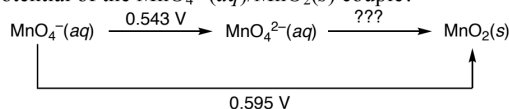


# Oxidation - Reduction

1. NO disproportionates into  $\text{N}_2\text{O}$  and  $\text{NO}_2^-$  in basic solution. When the equation for this reaction is balanced using smallest whole number coefficients, what is the coefficient of NO?

(A) 2      (B) 3      (C) 4      (D) 5

2. Given the standard reduction potentials in basic solution shown below, what is the value of the standard reduction potential of the  $\text{MnO}_4^{2-}(\text{aq})/\text{MnO}_2(\text{s})$  couple?



(A) 0.052 V      (B) 0.569 V  
(C) 0.621 V      (D) 0.647 V

3. For the voltaic cell  $\text{Cr}|\text{Cr}^{3+}(1.0 \text{ M})||\text{Ag}^+(1.0 \text{ M})|\text{Ag}$ , which change will cause the cell potential to decrease?

(A) Decreasing the pH of the silver half-cell by adding concentrated nitric acid (assuming a negligible change in volume)  
(B) Bubbling  $\text{NH}_3(\text{g})$  through the silver half-cell (assuming a negligible change in volume)  
(C) Adding water to the chromium half-cell  
(D) Increasing the size of the silver electrode

4. What is the molar solubility of lead sulfate at 298 K?

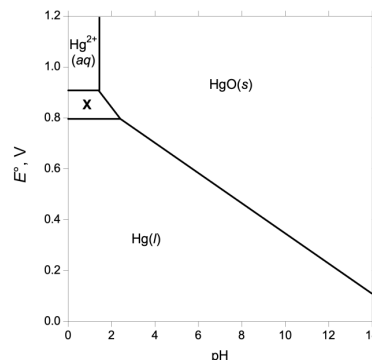
| Half-reaction                                                                                   | $E^\circ$ , V |
|-------------------------------------------------------------------------------------------------|---------------|
| $\text{Pb}^{2+}(\text{aq}) + 2 e^- \rightarrow \text{Pb}(\text{s})$                             | -0.126        |
| $\text{PbSO}_4(\text{s}) + 2 e^- \rightarrow \text{Pb}(\text{s}) + \text{SO}_4^{2-}(\text{aq})$ | -0.359        |

(A)  $1.31 \times 10^{-8} \text{ mol L}^{-1}$       (B)  $8.46 \times 10^{-7} \text{ mol L}^{-1}$   
(C)  $5.46 \times 10^{-5} \text{ mol L}^{-1}$       (D)  $1.14 \times 10^{-4} \text{ mol L}^{-1}$

5. A solution containing a metal ion is electrolyzed, with 1.50 g of the metal being deposited at the cathode. Concurrently, 0.265 L of dry oxygen gas is collected at the anode (measured at STP). What is the metal?

(A) Al      (B) Cu      (C) Sn      (D) U

6. A diagram showing the thermodynamic stability of mercury-containing species as a function of pH and reduction potential (Pourbaix diagram) is shown below. What is X?



(A)  $\text{Hg}_2^{2+}(\text{aq})$       (B)  $\text{Hg}_2\text{O}(\text{s})$   
(C)  $\text{Hg}(\text{OH})^+(\text{aq})$       (D)  $\text{Hg}(\text{O})(\text{OH})(\text{s})$

7. What is the average oxidation state of molybdenum in the phosphomolybdate ion  $\text{PMo}_{12}\text{O}_{40}^{3-}$ ?

(A) +6.00      (B) +6.17  
(C) +6.42      (D) +6.50

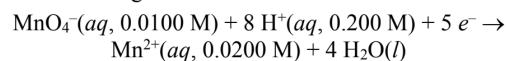
8. For a voltaic cell  $\text{Zn}|\text{Zn}^{2+}(1.0 \text{ M})||\text{Ag}^+(1.0 \text{ M})|\text{Ag}$ , which is correct?

(A) The zinc electrode is the anode.  
(B) The concentration of  $\text{Ag}^+$  increases as the cell operates.  
(C) The mass of the zinc electrode increases as the cell operates.  
(D) Electrons flow through the external circuit from the silver electrode to the zinc electrode.

9. A 1.000 g sample of a gold compound is electrolyzed exhaustively to deposit all the gold present on the cathode. This requires 4700 s of electrolysis with a constant current of 0.130 A. What is the compound?

(A)  $\text{LiAuCl}_2$  ( $M = 274.8$ )      (B)  $\text{LiAuCl}_4$  ( $M = 345.7$ )  
(C)  $\text{CsAuCl}_2$  ( $M = 400.8$ )      (D)  $\text{CsAuCl}_4$  ( $M = 471.7$ )

10. The standard reduction potential  $E^\circ$  for the reduction of permanganate to manganese(II) in acidic solution is 1.51 V at 298 K. What is the value of  $E$  for the half-reaction at 298 K at the given concentrations?



(A) 1.16 V      (B) 1.44 V  
(C) 1.50 V      (D) 1.51 V

# Oxidation - Reduction

- $$\text{HSeO}_4^-(aq) + 9 \text{H}^+(aq) + 8 e^- \rightarrow \text{H}_2\text{Se}(aq) + 4 \text{H}_2\text{O}(l)$$
- $$E^\circ = ???$$

(A) 0.30 V                      (B) 0.63 V  
(C) 0.71 V                      (D) 1.78 V

- (A)  $\text{NaBO}_3$                       (B)  $\text{NaNO}_3$   
(C)  $\text{K}_3\text{PO}_4$                     (D)  $\text{K}_2\text{Cr}_2\text{O}_7$

- $$\text{Cu}(s) + \text{H}^+(aq) + \text{NO}_3^-(aq) \rightarrow \text{Cu}^{2+}(aq) + \text{NO}(g) + \text{H}_2\text{O}(l)$$

14. In the Hall-Héroult process for producing aluminum, aluminum is electrolytically reduced from  $\text{Al}_2\text{O}_3$  dissolved in molten cryolite,  $\text{Na}_3\text{AlF}_6$ , while a graphite anode is oxidized to  $\text{CO}_2$ . What is the ratio of the mass of Al deposited on the cathode to the mass of C lost from the anode?

15. The cathode of an electrolytic cell with a copper electrode initially has 100.0 mL of a solution that is 1.00 M in both  $\text{Cu}^{2+}(aq)$  and  $\text{Fe}(\text{CN})_6^{3-}(aq)$ . The solution is electrolyzed with a constant current of 0.500 A at a constant temperature of 298 K. How much time elapses before  $\text{Cu}(s)$  begins to deposit on the electrode?

(A) 0 min                      (B) 89.4 min  
(C) 252 min                  (D) 322 min

- (A) The excited state is both a stronger oxidizing agent and a stronger reducing agent than the ground state.
- (B) The excited state is both a weaker oxidizing agent and a weaker reducing agent than the ground state.
- (C) The excited state is a stronger oxidizing agent but a weaker reducing agent than the ground state.
- (D) The excited state is a weaker oxidizing agent but a stronger reducing agent than the ground state.

- $$2 \text{ PbO}(s) \rightarrow \text{PbO}_2(s) + \text{Pb}(s) \quad \Delta G^\circ = ???$$

(A)  $163 \text{ kJ mol}^{-1}$                       (B)  $241 \text{ kJ mol}^{-1}$   
(C)  $403 \text{ kJ mol}^{-1}$                       (D)  $483 \text{ kJ mol}^{-1}$

- $$\underline{\hspace{1cm}} \text{P}_4(s) + \underline{\hspace{1cm}} \text{OH}^-(aq) + \underline{\hspace{1cm}} \text{H}_2\text{O}(l) \rightarrow \underline{\hspace{1cm}} \text{PH}_3(g) + \underline{\hspace{1cm}} \text{PO}_4^{3-}(aq)$$

- (A)  $\text{P}_4(s)$                       (B)  $\text{OH}^-(aq)$   
(C)  $\text{H}_2\text{O}(l)$                     (D)  $\text{PH}_3(g)$

- (A)  $\text{H}_2(g)$                       (B)  $\text{Ni}(s)$   
(C)  $\text{Sn}^{2+}(aq)$                   (D)  $\text{Zn}(s)$

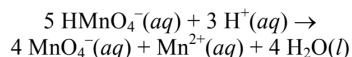
- 2

# Oxidation - Reduction

21. The cathodic compartment of an electrolytic cell is 0.100 M in the ions  $\text{Fe}(\text{CN})_6^{3-}$  and  $\text{Cu}^{2+}$  and has a chemically inert electrode. As current is passed through the cell, which best describes how  $\text{Cu}(\text{s})$  is deposited on the electrode?

| Half-reaction                                                               | $E^\circ$ , V |
|-----------------------------------------------------------------------------|---------------|
| $\text{Cu}^{2+}(\text{aq}) + 2e^- \rightarrow \text{Cu}(\text{s})$          | +0.337        |
| $\text{Fe}(\text{CN})_6^{3-} + e^- \rightarrow \text{Fe}(\text{CN})_6^{4-}$ | +0.370        |

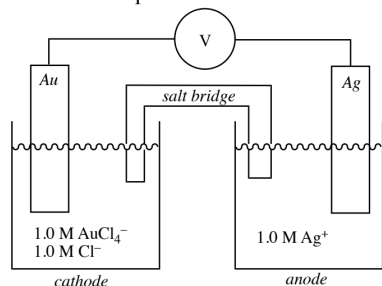
- (A) Copper is deposited immediately, but at a rate much lower than 1 mol per 193000 C. As the electrolysis proceeds, the rate of copper deposition increases.
- (B) Copper is deposited immediately, at a rate close to 1 mol per 193000 C. As the electrolysis proceeds, the rate of copper deposition decreases.
- (C) No copper is deposited for a certain length of time, then copper deposition begins.
- (D) Copper is deposited at a rate of 1 mol per 193000 C for a certain length of time, then the rate of copper deposition decreases.
22. What is  $E^\circ$  for the disproportionation of hydrogen manganate ion in acidic solution?



| Half-reaction                                                                                                                        | $E^\circ$ , V |
|--------------------------------------------------------------------------------------------------------------------------------------|---------------|
| $\text{MnO}_4^-(\text{aq}) + \text{H}^+(\text{aq}) + e^- \rightarrow \text{HMnO}_4^-(\text{aq})$                                     | +0.90         |
| $\text{MnO}_4^-(\text{aq}) + 8 \text{H}^+(\text{aq}) + 5 e^- \rightarrow \text{Mn}^{2+}(\text{aq}) + 4 \text{H}_2\text{O}(\text{l})$ | +1.51         |

- (A) 0.61 V (B) 0.76 V (C) 0.99 V (D) 1.21 V

23. In the galvanic cell shown, the silver electrode is oxidized to  $\text{Ag}^+$  while  $\text{AuCl}_4^-$  is reduced to elemental Au, and the measured cell potential is 0.13 V.



Which change will result in the largest value of the measured cell potential?

- (A) The concentration of  $\text{AuCl}_4^-$  is decreased to 0.10 M.
- (B) The concentration of  $\text{Cl}^-$  is decreased to 0.10 M.
- (C) The concentration of  $\text{Ag}^+$  is decreased to 0.10 M.
- (D) The surface area of the Ag electrode is decreased by a factor of 10.

24. In which species does the underlined element have the highest oxidation state?

- (A)  $\text{H}\underline{\text{P}}\text{O}_4^{2-}$  (B)  $\underline{\text{Mn}}\text{O}_4^{2-}$   
(C)  $\text{H}\underline{\text{S}}\text{O}_3^-$  (D)  $\underline{\text{N}}\text{O}_3^-$

25. Oxalic acid reacts with acidified potassium permanganate according to the unbalanced equation below. What is the coefficient for  $\text{H}_2\text{O}$  when the equation is balanced with the smallest whole number coefficients?

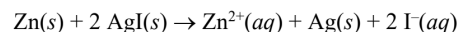


- (A) 2 (B) 3 (C) 5 (D) 8

26. An aqueous solution containing both  $\text{Cu}(\text{II})$  ions and  $\text{Rh}(\text{III})$  ions is electrolyzed. After passage of 0.153 A for 16400 s, 0.826 g of metal has been deposited. What is the composition of the deposited metal?

- (A) Cu only  
(B) Rh only  
(C) A mixture of Cu and Rh  
(D) It cannot be determined from the information given.

27. Which statements about the following cell are correct?

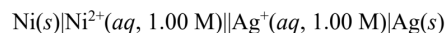


| Species                     | $\Delta H_f^\circ$ , kJ mol <sup>-1</sup> | $S^\circ$ , J mol <sup>-1</sup> K <sup>-1</sup> |
|-----------------------------|-------------------------------------------|-------------------------------------------------|
| $\text{Zn}(\text{s})$       | 0                                         | 41.6                                            |
| $\text{AgI}(\text{s})$      | -61.84                                    | 115.5                                           |
| $\text{Zn}^{2+}(\text{aq})$ | -153.89                                   | -112.1                                          |
| $\text{Ag}(\text{s})$       | 0                                         | 42.6                                            |
| $\text{I}^-(\text{aq})$     | -55.19                                    | 111.3                                           |

- I. The standard cell potential at 298 K is 0.54 V.
- II. The cell potential increases with increasing temperature.

- (A) I only (B) II only  
(C) Both I and II (D) Neither I nor II

28. The cell potential of the following at 298 K is 1.05 V.



Sodium cyanide is added to each half-cell until the concentration of free  $\text{CN}^-$  reaches 1.00 M. What is the new observed cell potential? The  $K_f$  of  $\text{Ni}(\text{CN})_4^{2-}$  is  $2.0 \times 10^{31}$  and the  $K_f$  of  $\text{Ag}(\text{CN})_2^-$  is  $9.8 \times 10^{21}$ .

- (A) 0.68 V (B) 0.78 V (C) 1.33 V (D) 1.43 V

# Oxidation - Reduction

29. What is the standard reduction potential of hydrogen selenate ion to form elemental selenium under acidic conditions at 298 K?

| Half-reaction                                                                                                  | $E^\circ$ , V |
|----------------------------------------------------------------------------------------------------------------|---------------|
| $\text{H}_2\text{SeO}_3(aq) + 4 \text{H}^+(aq) + 4e^- \rightarrow \text{Se}(s) + 3 \text{H}_2\text{O}(l)$      | 0.74          |
| $\text{HSeO}_4^-(aq) + 3 \text{H}^+(aq) + 2e^- \rightarrow \text{H}_2\text{SeO}_3(aq) + \text{H}_2\text{O}(l)$ | 1.15          |
| $\text{HSeO}_4^-(aq) + 7 \text{H}^+(aq) + 6e^- \rightarrow \text{Se}(s) + 4 \text{H}_2\text{O}(l)$             | ???           |

(A) 0.88 V (B) 0.94 V (C) 1.56 V (D) 1.89 V

30. In basic solution,  $\text{Cl}_2(g)$  disproportionates to form  $\text{Cl}^-(aq)$  and  $\text{ClO}_3^-(aq)$ . What is the mole ratio of  $\text{Cl}^-(aq)$  to  $\text{ClO}_3^-(aq)$  produced in this reaction?

(A) 5:1 (B) 4:1 (C) 3:1 (D) 3:2

31. What are the oxidation states of the sulfur atoms in the thiosulfate ion,  $\text{S}_2\text{O}_3^{2-}$ ?

(A) Both sulfur atoms have an oxidation state of +2.  
 (B) Both sulfur atoms have an oxidation state of +3.  
 (C) One sulfur atom has an oxidation state of +5 and one has an oxidation state of -1.  
 (D) One sulfur atom has an oxidation state of +6 and one has an oxidation state of -2.

32. A new electrode material is being tested for its ability to selectively reduce oxygen gas to hydrogen peroxide. After running an electrolytic cell using this electrode for 110.0 minutes at a current of 0.150 A, the solution in the cathode compartment is analyzed and is found to contain  $2.00 \times 10^{-3} \text{ mol H}_2\text{O}_2$ . What is the percent yield of  $\text{H}_2\text{O}_2$  based on the current passed through the cell (the faradaic yield)?

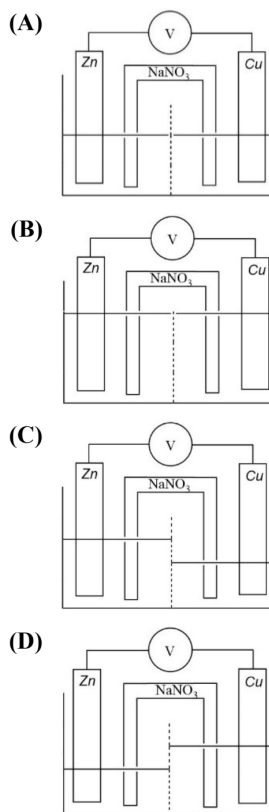
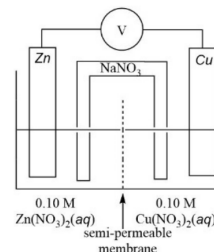
(A) 19.5% (B) 39.0% (C) 78.0% (D) 100%

33. What is the  $K_{sp}$  of  $\text{AgI}$  at 298 K?

| Half-reaction                                                   | $E^\circ$ , V |
|-----------------------------------------------------------------|---------------|
| $\text{Ag}^+(aq) + e^- \rightarrow \text{Ag}(s)$                | +0.80         |
| $\text{AgI}(s) + e^- \rightarrow \text{Ag}(s) + \text{I}^-(aq)$ | -0.15         |

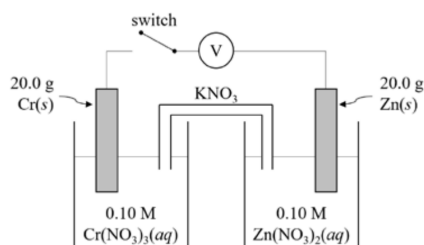
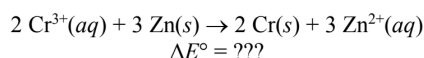
(A)  $4.8 \times 10^{-26}$  (B)  $1.0 \times 10^{-22}$   
 (C)  $8.5 \times 10^{-17}$  (D)  $1.0 \times 10^{-11}$

34. In the diagram below, an electrochemical cell's anode and cathode are linked by a semi-permeable membrane that allows only water to pass. The concentrations of  $\text{Zn}(\text{NO}_3)_2(aq)$  and  $\text{Cu}(\text{NO}_3)_2(aq)$  are initially each 0.10 M. Which of the following diagrams best shows the water levels in each cell at equilibrium?



# Oxidation - Reduction

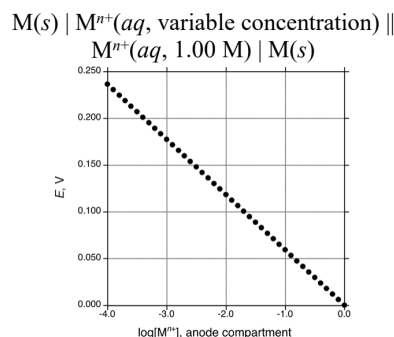
35. Consider the voltaic cell below at 25 °C with 1.00 L of solution in each half-cell. When the switch is closed, the mass of the Cr(s) electrode increases until the cell is fully discharged (i.e.,  $\Delta E = 0$  V). If the final mass of Cr(s) is 24.50 g, what is  $\Delta E^\circ$  at 25 °C for the reaction shown below?



- (A) 0.012 V      (B) 0.018 V  
(C) 0.024 V      (D) 0.054 V
36. When the following reaction is balanced under basic conditions, what is the ratio of the coefficients of  $\text{Mn}(\text{OH})_2(s)$  to  $\text{MnO}_4^{2-}(aq)$ ?
- $$\text{Mn}(\text{OH})_2(s) + \text{MnO}_4^-(aq) \rightarrow \text{MnO}_4^{2-}(aq)$$
- (A) 3 : 1      (B) 1 : 3      (C) 1 : 4      (D) 1 : 5
37. What is the standard reduction potential for the reduction of permanganate ion to manganese dioxide in acidic solution?
- | Half-Reaction                                                                                            | $E^\circ$ , V |
|----------------------------------------------------------------------------------------------------------|---------------|
| $\text{MnO}_4^-(aq) + 8 \text{H}^+(aq) + 5 e^- \rightarrow \text{Mn}^{2+}(aq) + 4 \text{H}_2\text{O}(l)$ | +1.51         |
| $\text{MnO}_2(s) + 4 \text{H}^+(aq) + 2 e^- \rightarrow \text{Mn}^{2+}(aq) + 2 \text{H}_2\text{O}(l)$    | +1.23         |
| $\text{MnO}_4^-(aq) + 4 \text{H}^+(aq) + 3 e^- \rightarrow \text{MnO}_2(s) + 2 \text{H}_2\text{O}(l)$    | ???           |
- (A) +0.28 V      (B) +1.37 V  
(C) +1.43 V      (D) +1.70 V
38. A current of 1.20 A passed through a solution of a metal nitrate for 153 minutes plates out 3.92 g of the metal. What is the metal?
- (A) Ni      (B) Cu      (C) Rh      (D) Ba
39. Which changes will lead to an increase in the theoretical cell potential for a hydrogen-oxygen fuel cell?
- A more efficient catalyst is used at the cathode.
  - The temperature is increased from 298 K to 320 K.
- (A) I only      (B) II only  
(C) Both I and II      (D) Neither I nor II

40. Metallic gold will not dissolve in nitric acid alone, but will dissolve in a mixture of nitric and hydrochloric acids. Which is the best explanation for this observation?
- (A) Chloride ion complexes gold(III), stabilizing it as  $\text{AuCl}_4^-$ .  
(B) Nitric acid is a stronger oxidant in more acidic solution.  
(C) Nitric and hydrochloric acids react to form  $\text{NO}_2\text{Cl}$ , which reacts readily with metallic gold.  
(D) Nitric and hydrochloric acids react to form  $\text{Cl}_2$ , which reacts readily with metallic gold.

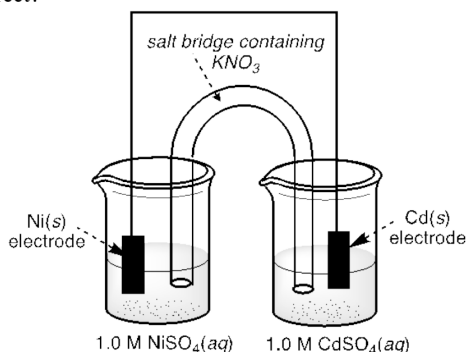
41. A concentration cell is set up with a metal M and aqueous solutions of the  $\text{M}^{n+}$  ion. The cell potential at 298 K as a function of the logarithm of the  $[\text{M}^{n+}]$  in the anode is graphed below. What may be concluded about the value of  $n$ ?



- (A)  $n = 1$   
(B)  $n = 2$   
(C)  $n = 3$   
(D)  $n$  cannot be determined from the information given.
42. The cathode of an electrolytic cell is placed in 100 mL of a solution that is 0.01 M in both  $\text{AgNO}_3$  and  $\text{Cu}(\text{NO}_3)_2$ . A current of 0.5 A is passed through the cell for 250 s. What deposits on the cathode?
- (A) Ag only  
(B) Cu only  
(C)  $1 \times 10^{-3}$  mol Ag and some of the Cu  
(D)  $1 \times 10^{-3}$  mol Cu and some of the Ag
43. What is the oxidation state of sulfur in  $\text{Na}_2\text{S}_2\text{O}_4$ ?
- (A) -2      (B) +3      (C) +4      (D) +6

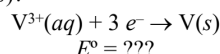
# Oxidation - Reduction

44. Which statement about the electrochemical cell is correct?



| Half-Reaction                                                       | $E^\circ$ , V |
|---------------------------------------------------------------------|---------------|
| $\text{Cd}^{2+}(\text{aq}) + 2 e^- \rightarrow \text{Cd}(\text{s})$ | -0.40         |
| $\text{Ni}^{2+}(\text{aq}) + 2 e^- \rightarrow \text{Ni}(\text{s})$ | -0.25         |

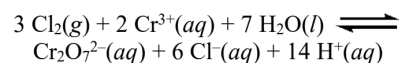
- (A) Ni(s) is the anode.  
 (B) As the cell discharges,  $\text{K}^+$  ions migrate toward the  $\text{NiSO}_4$  solution.  
 (C) Using a thinner Ni(s) electrode would increase the magnitude of the cell potential.  
 (D) Adding  $\text{NiSO}_4$  to the beaker on the left would decrease the magnitude of the cell potential.
45. What is the standard reduction potential for the reduction of  $\text{V}^{3+}(\text{aq})$  to  $\text{V}(\text{s})$ ?



| Half-Reaction                                                         | $E^\circ$ , V |
|-----------------------------------------------------------------------|---------------|
| $\text{V}^{2+}(\text{aq}) + 2 e^- \rightarrow \text{V}(\text{s})$     | -1.13 V       |
| $\text{V}^{3+}(\text{aq}) + e^- \rightarrow \text{V}^{2+}(\text{aq})$ | -0.26 V       |

- (A) -0.70 V      (B) -0.84 V  
 (C) -1.39 V      (D) -1.65 V
46. For the cell,
- $$\text{Ag}(\text{s}) \mid \text{Ag}(\text{NH}_3)_2^+(\text{aq}, 0.010 \text{ M}) \mid \text{NH}_3(\text{aq}, 0.40 \text{ M}) \parallel \text{Ag}^+(\text{aq}, 0.010 \text{ M}) \mid \text{Ag}(\text{s})$$
- What is the cell potential? The  $K_f$  of  $\text{Ag}(\text{NH}_3)_2^+$  is  $1.6 \times 10^7$ .
- (A) 0.12 V    (B) 0.38 V    (C) 0.40 V    (D) 0.44 V

47. What is the equilibrium constant for the reaction at 298 K?



| Half-Reaction                                                                                                                             | $E^\circ$ , V |
|-------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| $\text{Cr}_2\text{O}_7^{2-}(\text{aq}) + 14 \text{H}^+(\text{aq}) + 6 e^- \rightarrow 2 \text{Cr}^{3+}(\text{aq}) + 7 \text{H}_2\text{O}$ | 1.33          |
| $\text{Cl}_2(\text{g}) + 2 e^- \rightarrow 2 \text{Cl}^-(\text{aq})$                                                                      | 1.36          |

- (A) 3.2      (B) 1100  
 (C)  $3.2 \times 10^{46}$       (D)  $1.1 \times 10^{279}$
48. Iodide ion is oxidized to hypoiodite ion by permanganate ion in basic solution according to the following unbalanced equation:
- $$\text{I}^-(\text{aq}) + \text{MnO}_4^-(\text{aq}) \rightarrow \text{IO}^-(\text{aq}) + \text{MnO}_2(\text{s})$$
- What is the ratio of hydroxide ions to iodide ions in the balanced equation?
- (A) 3 : 1    (B) 2 : 1    (C) 1 : 1    (D) 2 : 3
49. In the carbon monoxide complex  $\text{Na}[\text{V}(\text{CO})_6]$ , what is the oxidation number of vanadium?
- (A) -1    (B) +3    (C) +5    (D) +6
50. What is the  $K_{sp}$  of  $\text{Fe}(\text{OH})_2$  at 298 K?

| Half-Reaction                                                                                         | $E^\circ$ , V (at 298 K) |
|-------------------------------------------------------------------------------------------------------|--------------------------|
| $\text{Fe}^{2+}(\text{aq}) + 2 e^- \rightarrow \text{Fe}(\text{s})$                                   | -0.44                    |
| $\text{Fe}(\text{OH})_2(\text{s}) + 2 e^- \rightarrow \text{Fe}(\text{s}) + 2 \text{OH}^-(\text{aq})$ | -0.89                    |

- (A)  $2.4 \times 10^{-8}$       (B)  $6.0 \times 10^{-16}$   
 (C)  $1.0 \times 10^{-45}$       (D)  $8.3 \times 10^{-46}$
51. A 0.100-g sample of an alloy of copper and silver is dissolved in nitric acid to give a solution containing  $\text{Cu}^{2+}$  and  $\text{Ag}^+$  ions. This mixture is electrolyzed exhaustively to redeposit all the metal ions as the elemental metals, requiring a current of 0.150 A for 1429 s. What is the mass percent of copper in the alloy?
- (A) 20.0%    (B) 58.3%    (C) 70.6%    (D) 77.2%

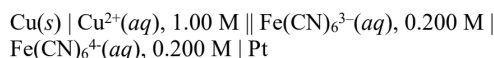
# Oxidation - Reduction

52. Which electrochemical cell has the greatest (most positive) standard cell potential?

| Half-Reaction                                                     | $E^\circ$ , V |
|-------------------------------------------------------------------|---------------|
| $\text{Cd}^{2+}(aq) + 2e^- \rightarrow \text{Cd}(s)$              | -0.40         |
| $\text{Tl}^+(aq) + e^- \rightarrow \text{Tl}(s)$                  | -0.34         |
| $\text{AgCl}(s) + e^- \rightarrow \text{Ag}(s) + \text{Cl}^-(aq)$ | +0.22         |
| $\text{Cu}^{2+}(aq) + 2e^- \rightarrow \text{Cu}(s)$              | +0.34         |

- (A)  $\text{Cu}(s) | \text{Cu}^{2+}(aq) || \text{Cd}^{2+}(aq) | \text{Cd}(s)$   
 (B)  $\text{Tl}(s) | \text{Tl}^+(aq) || \text{AgCl}(s) | \text{Ag}(s), \text{Cl}^-(aq)$   
 (C)  $\text{Cd}(s) | \text{Cd}^{2+}(aq) || \text{AgCl}(s) | \text{Ag}(s), \text{Cl}^-(aq)$   
 (D)  $\text{Tl}(s) | \text{Tl}^+(aq) || \text{Cu}^{2+}(aq) | \text{Cu}(s)$

53. The galvanic cell below has equal volumes in the two half-cells. As it is discharged at 25 °C, the cell potential gradually decreases to zero, whereupon the discharge stops. What is the concentration of  $\text{Cu}^{2+}(aq)$  when the discharge stops?



| Half-Reaction                                                                       | $E^\circ$ , V |
|-------------------------------------------------------------------------------------|---------------|
| $\text{Cu}^{2+}(aq) + 2e^- \rightarrow \text{Cu}(s)$                                | +0.337        |
| $\text{Fe}(\text{CN})_6^{3-}(aq) + e^- \rightarrow \text{Fe}(\text{CN})_6^{4-}(aq)$ | +0.360        |

- (A) 0.80 M (B) 0.96 M (C) 1.04 M (D) 1.20 M

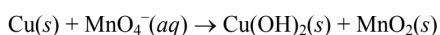
54. Which changes to this galvanic cell increase the measured potential?



- I. Increasing  $[\text{Cu}^{2+}]$  in the  $\text{Cu}/\text{Cu}^{2+}$  half-cell to 0.50 M  
 II. Adding  $\text{Cl}^-$  to the  $\text{Ag}^+/\text{Ag}$  half-cell until  $[\text{Cl}^-] = 0.01 \text{ M}$

- (A) I only (B) II only  
 (C) Both I and II (D) Neither I nor II

55. When the following skeleton equation is balanced with smallest whole number coefficients, what is the coefficient and location of  $\text{H}_2\text{O}(l)$ ? The reaction takes place in basic solution.



- (A) 2, on reactant side (B) 4, on reactant side  
 (C) 2, on product side (D) 4, on product side

56. Which of the following species contains the element in the highest oxidation state?

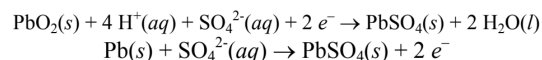
- (A)  $\text{OsO}_4$  (B)  $\text{Mn}_2(\text{CO})_{10}$   
 (C)  $\text{N}_5^+$  (D)  $\text{XeF}_8^{2-}$

57. What is the standard reduction potential of  $\text{Hg}_2^{2+}(aq)$  to  $\text{Hg}(l)$ ?

| Half Reaction                                                  | $E^\circ$ , V |
|----------------------------------------------------------------|---------------|
| $2 \text{Hg}^{2+}(aq) + 2e^- \rightarrow \text{Hg}_2^{2+}(aq)$ | +0.90         |
| $\text{Hg}_2^{2+}(aq) + 2e^- \rightarrow 2 \text{Hg}(l)$       | +0.80         |

- (A) +1.70 V (B) +0.85 V  
 (C) +0.10 V (D) -0.10 V

58. The lead-acid storage battery consists of the following two half-cells:

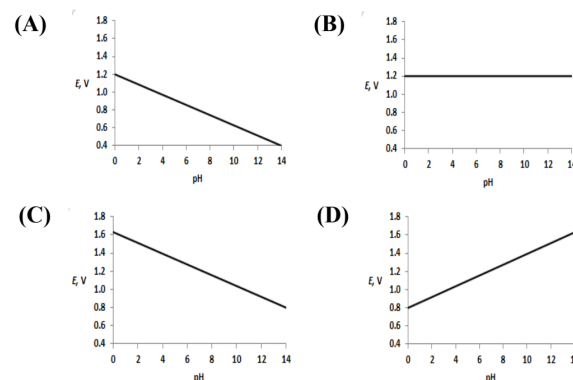
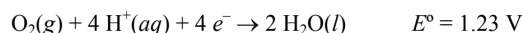


Which of the following concentrations decrease as the battery is discharged?

- I.  $[\text{H}^+]$  II.  $[\text{Pb}^{2+}]$

- (A) I only (B) II only  
 (C) Both I and II (D) Neither I nor II

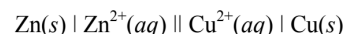
59. Which graph represents the reduction potential of  $\text{O}_2$  (at 1 bar pressure and 25 °C) as a function of pH?



60. What is the oxidation state of nitrogen in magnesium nitride?

- (A) -3 (B) -2 (C) +3 (D) +5

61. For the galvanic cell



the standard cell potential is 1.10 V. What is the potential at 25 °C if the concentrations of  $\text{Zn}^{2+}$  and  $\text{Cu}^{2+}$  are 2.5 M and 0.10 M respectively?

- (A) 1.14 V (B) 1.10 V (C) 1.06 V (D) 1.02 V

# Oxidation - Reduction

62. Which of the following metals is most likely to be produced through electrolysis?

(A) Aluminum (B) Gold  
(C) Iron (D) Mercury

63. What is the standard reduction potential for  $\text{Cr}^{2+}(\text{aq})$  to  $\text{Cr}(\text{s})$ ?

| Half-Reaction                                                           | $E^\circ, \text{V}$ |
|-------------------------------------------------------------------------|---------------------|
| $\text{Cr}^{3+}(\text{aq}) + 3e^- \rightarrow \text{Cr}(\text{s})$      | -0.73               |
| $\text{Cr}^{3+}(\text{aq}) + e^- \rightarrow \text{Cr}^{2+}(\text{aq})$ | -0.50               |

(A) -0.23 V (B) -0.85 V  
(C) -1.23 V (D) -1.69 V

64. What is observed if a piece of lead is placed in an aqueous solution of  $\text{Cu}(\text{NO}_3)_2$ ?

| Half-Reaction                                                      | $E^\circ, \text{V}$ |
|--------------------------------------------------------------------|---------------------|
| $\text{Pb}^{2+}(\text{aq}) + 2e^- \rightarrow \text{Pb}(\text{s})$ | -0.13               |
| $\text{Cu}^{2+}(\text{aq}) + 2e^- \rightarrow \text{Cu}(\text{s})$ | +0.34               |

- I. The mass of the metal piece increases.  
II. The blue color of the solution becomes less intense.

(A) I only (B) II only  
(C) Both I and II (D) Neither I nor II

65. If water is electrolyzed for 2.0 hr with a current of 10.0 A, what volume of dry oxygen gas is collected at STP?

(A) 4.2 L (B) 4.6 L (C) 8.4 L (D) 17 L

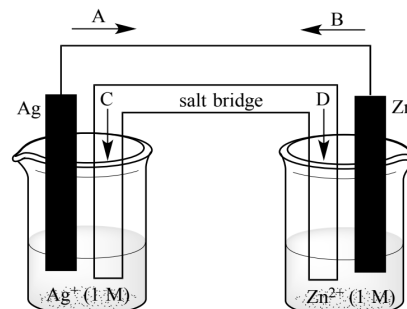
66. What is the oxidation number of Re in  $\text{Mg}(\text{ReO}_4)_2$ ?

(A) +4 (B) +5 (C) +6 (D) +7

67. It takes 126.5 minutes using a current of 5.15 A to deposit all of the nickel from 225 mL of a solution containing  $\text{Ni}^{2+}$ . What was the original concentration of  $\text{Ni}^{2+}$  in the solution?

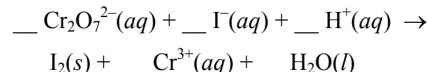
(A) 3.60 M (B) 1.80 M  
(C) 0.900 M (D)  $1.50 \times 10^{-2}$  M

68. In the galvanic cell shown below, which arrow indicates the spontaneous electron flow?



(A) A (B) B (C) C (D) D

69. What is the coefficient of  $\text{I}_2(\text{s})$  when the reaction below is balanced with smallest whole number coefficients?

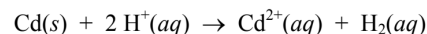


(A) 2 (B) 3 (C) 4 (D) 6

70. Lithium ion batteries are now commonly used in rechargeable consumer electronic devices. The main reason lithium is used in these devices is because

(A) lithium has a lower electronegativity than nickel in common nickel-cadmium batteries.  
(B) lithium batteries are not as toxic as common alkaline batteries.  
(C) lithium batteries have a reduced risk of leakage of chemicals.  
(D) lithium batteries achieve a greater amount of energy stored per unit mass than other common batteries.

71. The standard reduction potential of  $\text{Cd}^{2+}(\text{aq})$  is  $-0.402 \text{ V}$ . A voltaic cell described by



has  $[\text{Cd}^{2+}] = 0.900 \text{ M}$  and a hydrogen pressure of  $0.975 \text{ atm}$ . Its cell potential at  $25^\circ\text{C}$  is measured as  $E = +0.192 \text{ V}$ . What is the pH in the  $\text{H}^+ | \text{H}_2$  half-cell?

(A) 3.28 (B) 3.58 (C) 6.54 (D) 7.15

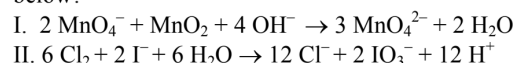
# Oxidation - Reduction

72. Based on the reactions and data below, which reaction is least spontaneous?

| Reduction                                                                    | $E^\circ$ (V) |
|------------------------------------------------------------------------------|---------------|
| $2 \text{Hg}^{2+} + 2 \text{e}^- \rightarrow \text{Hg}_2^{2+}$               | 1.82          |
| $\text{I}_2 + 2 \text{e}^- \rightarrow 2 \text{I}^-$                         | 0.53          |
| $2 \text{H}_2\text{O} + 2 \text{e}^- \rightarrow \text{H}_2 + 2 \text{OH}^-$ | -0.83         |

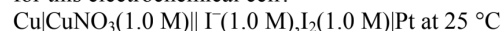
- (A)  $2 \text{Hg}^{2+} + 2 \text{I}^- \rightarrow \text{Hg}_2^{2+} + \text{I}_2$   
 (B)  $\text{H}_2 + 2 \text{OH}^- + \text{I}_2 \rightarrow 2 \text{H}_2\text{O} + 2 \text{I}^-$   
 (C)  $2 \text{H}_2\text{O} + \text{Hg}_2^{2+} \rightarrow \text{H}_2 + 2 \text{OH}^- + 2 \text{Hg}^{2+}$   
 (D)  $\text{H}_2 + 2 \text{OH}^- + 2 \text{Hg}^{2+} \rightarrow 2 \text{H}_2\text{O} + \text{Hg}_2^{2+}$

73. An increase in pH will promote which of the reactions below?



- (A) I only (B) II only  
 (C) Both I and II (D) Neither I nor II

74. Given the reactions and data below, what is the  $K$  value for this electrochemical cell?



| Reaction                                             | $E^\circ$ (V) |
|------------------------------------------------------|---------------|
| $\text{Cu}^+ + \text{e}^- \rightarrow \text{Cu}$     | 0.518         |
| $\text{I}_2 + 2 \text{e}^- \rightarrow 2 \text{I}^-$ | 0.534         |

- (A) 0.288 (B) 1.86 (C) 2.23 (D) 3.48

75. The advantages of methane fuel cells over internal combustion engines (ICEs) that burn methane include which of the following?

- I. Methane fuel cells are less polluting.  
 II. Methane fuel cells are more efficient.  
 III. Methane fuel cells are less expensive.

- (A) I only (B) I and II only  
 (C) I and III only (D) II and III only

76. A steady current of 1.20 Ampere is passed through a solution of  $\text{MCl}_x$  for 2 hours and 33 minutes. If 2.98 g of metal M are plated out, what is the identity of the metal?

- (A) Al (B) Cr (C) Ni (D) Zn

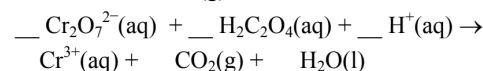
77.  $\text{MnO}_4^- + 5 \text{Fe}^{2+} + 8 \text{H}^+ \rightarrow \text{Mn}^{2+} + 5 \text{Fe}^{3+} + 4 \text{H}_2\text{O}$   
 $E^\circ = 0.743 \text{ V}$

What is the value of  $E$  for the cell based on the reaction above at  $25^\circ\text{C}$  for the following conditions?

| $[\text{MnO}_4^-]$             | $[\text{Fe}^{2+}]$           | $[\text{Mn}^{2+}]$ | $[\text{Fe}^{3+}]$ | pH   |
|--------------------------------|------------------------------|--------------------|--------------------|------|
| $2.3 \times 10^{-2} \text{ M}$ | $1 \times 10^{-4} \text{ M}$ | 1.0 M              | 1.0 M              | 3.00 |

- (A) 0.20 V (B) 0.45 V (C) 0.64 V (D) 1.28 V

78. When the equation below is balanced correctly using the simplest whole number coefficients, what is the coefficient for  $\text{CO}_2(\text{g})$ ?



- (A) 4 (B) 6 (C) 8 (D) 12

79.  $\text{Au}^{3+} + 3 \text{e}^- \rightarrow \text{Au}(\text{s}) \quad E^\circ = 1.50 \text{ V}$   
 $\text{Ag}^+ + \text{e}^- \rightarrow \text{Ag}(\text{s}) \quad E^\circ = 0.80 \text{ V}$   
 $\text{Cu}^{2+} + 2 \text{e}^- \rightarrow \text{Cu}(\text{s}) \quad E^\circ = 0.34 \text{ V}$

According to the standard reduction potentials above, a substance that can oxidize only one of these metals must have an  $E^\circ$  value

- (A) less than 0.34 V.  
 (B) between 0.34 and 0.80 V.  
 (C) between 0.80 and 1.50 V.  
 (D) greater than 1.50 V.

80. What is the  $\Delta G^\circ$  value for the electrochemical cell below?



| Half-Reaction                                                                                                                        | $E^\circ$ , V |
|--------------------------------------------------------------------------------------------------------------------------------------|---------------|
| $\text{NO}_3^-(\text{aq}) + 4 \text{H}^+(\text{aq}) + 3 \text{e}^- \rightarrow \text{NO}(\text{g}) + 2 \text{H}_2\text{O}(\text{l})$ | 0.960         |
| $\text{Cu}^{2+}(\text{aq}) + 2 \text{e}^- \rightarrow \text{Cu}(\text{s})$                                                           | 0.340         |

- (A) -753 kJ (B) -359 kJ  
 (C) -179 kJ (D) -59.8 kJ

81. An aqueous solution of  $\text{CuSO}_4$  is electrolyzed for 1.50 hours with a current of 2.50 amps. What mass of copper metal is formed?

- (A) 8.88 g (B) 4.44 g (C) 0.296 g (D) 0.0741 g

82. Under what conditions is the Nernst equation used to calculate cell potential voltages in a voltaic cell?

- (A) non-standard concentrations only  
 (B) non-spontaneous reactions only  
 (C) reactions at equilibrium only  
 (D) reactions of ions with the same charge only

| Half-Reaction                                                              | $E^\circ$ , V |
|----------------------------------------------------------------------------|---------------|
| $\text{Cu}^{2+}(\text{aq}) + 2 \text{e}^- \rightarrow \text{Cu}(\text{s})$ | 0.340         |
| $\text{Sn}^{2+}(\text{aq}) + 2 \text{e}^- \rightarrow \text{Sn}(\text{s})$ | -0.136        |
| $\text{Fe}^{2+}(\text{aq}) + 2 \text{e}^- \rightarrow \text{Fe}(\text{s})$ | -0.440        |
| $\text{Zn}^{2+}(\text{aq}) + 2 \text{e}^- \rightarrow \text{Zn}(\text{s})$ | -0.763        |
| $\text{Mg}^{2+}(\text{aq}) + 2 \text{e}^- \rightarrow \text{Mg}(\text{s})$ | -2.37         |

According to the standard reduction potentials above, cathodic protection of iron can be achieved by

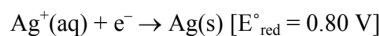
- (A)  $\text{Cu}(\text{s})$  only. (B)  $\text{Cu}(\text{s})$  and  $\text{Sn}(\text{s})$  only.  
 (C)  $\text{Mg}(\text{s})$  only. (D)  $\text{Mg}(\text{s})$  and  $\text{Zn}(\text{s})$  only.

# Oxidation - Reduction

84. Automobile catalytic converters are designed to
- (A) oxidize both CO and NO<sub>x</sub>.  
 (B) reduce both CO and NO<sub>x</sub>.  
 (C) oxidize CO and reduce NO<sub>x</sub>.  
 (D) reduce CO and oxidize NO<sub>x</sub>.
85. What is the average oxidation state of copper in the superconductor YBa<sub>2</sub>Cu<sub>3</sub>O<sub>7</sub>?
- (A) +2 (B) +2.33 (C) +2.67 (D) +3
86.  $\text{Sn}^{4+}(\text{aq}) + 2\text{e}^{-} \rightarrow \text{Sn}^{2+}(\text{aq}) \quad E^{\circ} = 0.15 \text{ V}$   
 $\text{Cr}^{3+}(\text{aq}) + \text{e}^{-} \rightarrow \text{Cr}^{2+}(\text{aq}) \quad E^{\circ} = -0.41 \text{ V}$   
 According to the standard reduction potentials above, what is the value of  $E^{\circ}$  for the reaction below?  
 $2\text{Cr}^{3+}(\text{aq}) + \text{Sn}^{2+}(\text{aq}) \rightarrow 2\text{Cr}^{2+}(\text{aq}) + \text{Sn}^{4+}(\text{aq})$
- (A) -0.97 V (B) -0.56 V  
 (C) +0.56 V (D) +0.97 V
87.  $\text{Ag}^{+}(\text{aq}) + \text{e}^{-} \rightarrow \text{Ag}(\text{s}) \quad E^{\circ} = 0.80 \text{ V}$   
 $\text{Mg}^{2+}(\text{aq}) + 2\text{e}^{-} \rightarrow \text{Mg}(\text{s}) \quad E^{\circ} = -2.73 \text{ V}$   
 Use the equations above to calculate the value of  $\Delta G^{\circ}$  (in kJ/mol) for the reaction:  
 $\text{Mg}(\text{s}) + 2\text{Ag}^{+}(\text{aq}) \rightarrow \text{Mg}^{2+}(\text{aq}) + 2\text{Ag}(\text{s})$
- (A) 681 (B) 341 (C) -341 (D) -681
88. What is the [Fe<sup>2+</sup>] in a cell at 25 °C for which  $E = -0.458 \text{ V}$  vs a standard hydrogen electrode?  
 $\text{Fe}^{2+}(\text{aq}) + 2\text{e}^{-} \rightarrow \text{Fe}(\text{s}) \quad E^{\circ} = -0.440 \text{ V}$
- (A) 0.246 M (B) 0.496 M  
 (C) 2.01 M (D) 4.06 M
89. Rechargeable batteries include which of those below?  
 I dry cell  
 II lead-acid storage battery  
 III nickel-cadmium battery
- (A) II only (B) I and II only  
 (C) II and III only (D) I, II, and III
90. How many liters of chlorine gas, Cl<sub>2</sub>, measured at 0 °C and 1 atm (STP) are released by the passage of 6.25 amperes for 1.85 hours through molten magnesium chloride?
- (A) 0.0805 L (B) 0.161 L  
 (C) 4.83 L (D) 9.67 L
91. What is the coefficient for Zn when the equation below is balanced with the smallest whole number coefficients?  
 $\text{Zn} + \text{H}^{+}(\text{aq}) + \text{NO}_3^{-}(\text{aq}) \rightarrow \text{Zn}^{2+}(\text{aq}) + \text{N}_2\text{O}(\text{g}) + \text{H}_2\text{O}(\text{l})$
- (A) 2 (B) 4 (C) 6 (D) 8
92. Which change occurs as the chemical reaction takes place in the standard electrochemical cell represented below?
- $\text{Zn}(\text{s})|\text{Zn}^{2+}(\text{aq})||\text{Sn}^{2+}(\text{aq})|\text{Sn}(\text{s})$
- | Reduction Half-reaction                                                     | $E^{\circ}$ (Volts at 298 K) |
|-----------------------------------------------------------------------------|------------------------------|
| $\text{Sn}^{2+}(\text{aq}) + 2\text{e}^{-} \rightarrow \text{Sn}(\text{s})$ | -0.136                       |
| $\text{Zn}^{2+}(\text{aq}) + 2\text{e}^{-} \rightarrow \text{Zn}(\text{s})$ | -0.763                       |
- I Electrons move through the external circuit from Zn to Sn.  
 II The concentration of Zn<sup>2+</sup>(aq) increases.  
 III The voltage increases from a negative value to zero.
- (A) I and II only (B) I and III only  
 (C) II and III only (D) I, II, and III
93. For the reaction:  
 $\text{C}_3\text{H}_8(\text{g}) + 5 \text{O}_2(\text{g}) \rightarrow 3 \text{CO}_2(\text{g}) + 4 \text{H}_2\text{O}(\text{l})$   
 $\Delta G^{\circ} = -2.108 \times 10^3 \text{ kJ}\cdot\text{mol}^{-1}$ . What is the value of the standard electrode potential,  $E^{\circ}$  for a fuel cell based on this reaction?
- (A) 1.09 V (B) 2.18 V (C) 4.37 V (D) 21.8 V
94. What occurs when an aqueous solution of Na<sub>2</sub>SO<sub>4</sub> containing several drops of phenolphthalein is electrolyzed between Pt electrodes?
- (A) The colorless solution turns pink at the anode but remains colorless at the cathode.  
 (B) The colorless solution turns pink at the cathode but remains colorless at the anode.  
 (C) The pink solution becomes colorless at the anode but remains pink at the cathode.  
 (D) The pink solution becomes colorless at the cathode but remains pink at the anode.
95. A current of 12 A is used to plate nickel from a Ni(NO<sub>3</sub>)<sub>2</sub> solution. Both Ni(s) and H<sub>2</sub>(g) are produced at the cathode. If the current efficiency with respect to the formation of Ni(s) is 62%, how many grams of nickel are plated on the cathode in 45 minutes?
- (A) 0.10 (B) 6.1 (C) 9.9 (D) 12
96. Calculate the cell potential,  $E$ , for a silver-silver chloride electrode immersed in 0.800 M KCl at 25 °C.  
 $[\text{Ag}^{+} + \text{e}^{-} \rightarrow \text{Ag}, E^{\circ} = 0.799 \text{ V}; K_{\text{sp}} = 1.8 \times 10^{-10}]$
- (A) 1.37 V (B) 0.80 V (C) 0.57 V (D) 0.23 V

# Oxidation - Reduction

97. An unknown metal, M, and its salt,  $M(NO_3)_2$ , are combined with a half-cell in which the following reaction occurs:



If  $E^\circ_{cell} = 1.36 \text{ V}$ , what is  $E^\circ_{red}$  for  $M^{2+}(aq) + 2e^- \rightarrow M(s)$ ?

- (A) 0.56 V (B) 0.24V (C) -0.24V (D) -0.56V

98. Given the standard reduction potentials:

|                                       |                             |
|---------------------------------------|-----------------------------|
| $O_2 + 4H^+ + 4e^- \rightarrow 2H_2O$ | $E^\circ = 1.23 \text{ V}$  |
| $Br_2 + 2e^- \rightarrow 2Br^-$       | $E^\circ = 1.08 \text{ V}$  |
| $2H^+ + 2e^- \rightarrow H_2$         | $E^\circ = 0.00 \text{ V}$  |
| $Na^+ + e^- \rightarrow Na$           | $E^\circ = -2.71 \text{ V}$ |

What products are formed in the electrolysis of 1 M NaBr in a solution with  $[H_3O^+] = 1 \text{ M}$ ?

- (A) Na(s) and  $O_2(g)$  (B) Na(s) and  $Br_2(g)$   
(C)  $H_2(g)$  and  $Br_2(g)$  (D)  $H_2(g)$  and  $O_2(g)$

99. According to the standard reduction potentials:

|                                        |                             |
|----------------------------------------|-----------------------------|
| $Pb^{2+}(aq) + 2e^- \rightarrow Pb(s)$ | $E^\circ = -0.13 \text{ V}$ |
| $Fe^{2+}(aq) + 2e^- \rightarrow Fe(s)$ | $E^\circ = -0.44 \text{ V}$ |
| $Zn^{2+}(aq) + 2e^- \rightarrow Zn(s)$ | $E^\circ = -0.76 \text{ V}$ |

Which species will reduce  $Mn^{3+}$  to  $Mn^{2+}$  [ $E^\circ = 1.51 \text{ V}$ ] but will NOT reduce  $Cr^{3+}$  to  $Cr^{2+}$  [ $E^\circ = -0.40 \text{ V}$ ]?

- (A) Pb only (B) Zn only  
(C) Pb and Fe only (D) Pb, Fe, and Zn

100.  $Zn(s) / Zn^{2+}(aq) // H^+(aq) / H_2(g)$   $E^\circ = 0.76 \text{ V}$   
What must be the pH in the hydrogen compartment of the cell designated above if the cell voltage is 0.70 V?  
(Assume that both the  $[Zn^{2+}]$  and the  $H_2(g)$  pressure are at standard values and  $T = 25^\circ \text{C}$ .)  
(A) 0.51 (B) 1.01 (C) 2.50 (D) 3.21

101. The equilibrium constant,  $K$ , is  $2.0 \times 10^{19}$  for the cell  
 $Ni(s) / Ni^{2+}(aq) // Hg_2^{2+}(aq) / Hg(l)$   
The value of  $E^\circ$  at  $25^\circ \text{C}$  for this cell is closest to  
(A) -1.14V (B) -0.57V (C) 0.57V (D) 1.14V

102. In a battery with a zinc anode, what is the minimum mass of zinc required if a current of 250 mA is drawn for 12.0 minutes?  
(A) 0.0610 g (B) 0.122 g  
(C) 0.244 g (D) 1.02 g

103. When connected to a Standard Hydrogen Electrode (SHE) electrons flow from an unknown half cell to the SHE. Which statement is correct?

- (A) The unknown half cell is the anode.  
(B) Oxidation occurs at the SHE.  
(C)  $E^\circ_{red}$  for the unknown half cell is positive.  
(D)  $E^\circ_{cell}$  is negative.

104. Given these standard

|                                             |                             |
|---------------------------------------------|-----------------------------|
| $Co^{3+}(aq) + e^- \rightarrow Co^{2+}(aq)$ | $E^\circ = 1.82 \text{ V}$  |
| $Co^{2+}(aq) + 2e^- \rightarrow Co(s)$      | $E^\circ = -0.28 \text{ V}$ |

reduction potentials, what is the standard reduction potential for  $Co^{3+}(aq) + 3e^- \rightarrow Co(s)$ ?

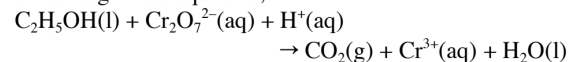
- (A) 2.10 V (B) 1.54 V  
(C) 1.26 V (D) 0.42 V

105. The cell

$Al(s) | Al^{3+}(aq, 0.001 \text{ M}) || Cu^{2+}(aq, 0.10 \text{ M}) | Cu(s)$   
has a standard cell potential,  $E^\circ = 2.00 \text{ V}$ . What is the cell potential for this cell at the concentrations given?

- (A) 2.07 V (B) 2.03 V (C) 1.97 V (D) 1.94 V

106. Ethanol reacts with dichromate ions in acid solution according to the equation;



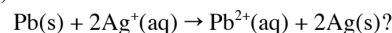
What is the coefficient for  $H^+(aq)$  when this equation is balanced with the smallest whole number coefficients?

- (A) 10 (B) 12 (C) 14 (D) 16

107. Given these standard

|                                        |                             |
|----------------------------------------|-----------------------------|
| $Ag^+(aq) + e^- \rightarrow Ag(s)$     | $E^\circ = 0.80 \text{ V}$  |
| $Pb^{2+}(aq) + 2e^- \rightarrow Pb(s)$ | $E^\circ = -0.13 \text{ V}$ |

reduction potentials, what is the free energy change (in  $\text{kJ} \cdot \text{mol}^{-1}$ ) for the reaction



- (A) -180 (B) -90 (C) 90 (D) 180

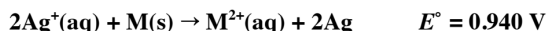
108. Chromium metal can be produced by the electrolysis of molten  $CrO_3$ . What current in amperes operating for 100 minutes is needed to produce 104 grams of this metal?

- (A) 193 (B) 96.5 (C) 64.3 (D) 32.2

# Oxidation - Reduction

109. In an operating voltaic cell electrons move through the external circuit and ions move through the electrolyte solution. Which statement describes these movements?
- (A) Electrons and negative ions both move toward the anode.  
 (B) Electrons and negative ions both move toward the cathode.  
 (C) Electrons move toward the anode and negative ions move toward the cathode.  
 (D) Electrons move toward the cathode and negative ions move toward the anode.
110. The reduction potentials for the  $+2$  cations, e.g.  $A^{2+} + 2e^- \rightarrow A^0$ , of four metals decrease in the order A, B, C, D. Which statement(s) is/are true?
- |                             |
|-----------------------------|
| I. $A^0$ reduces $B^{2+}$   |
| II. $B^{2+}$ oxidizes $C^0$ |
| III. $B^0$ oxidizes $D^0$   |
- (A) II only                      (B) III only  
 (C) I and II only              (D) I and III only

Questions 111 and 112 should be answered with reference to the reaction:



111. What is the value of  $E^\circ$  for the half reaction,  $M^{2+}(aq) + 2e^- \rightarrow M(s)$ ?
- |                                                |
|------------------------------------------------|
| $E^\circ / \text{V}$                           |
| $Ag^+(aq) + e^- \rightarrow Ag(s) \quad 0.799$ |
- (A) 0.658 V                      (B) 0.141 V  
 (C) -0.141 V                    (D) -0.658 V
112. Which change will cause the largest increase in the voltage of a cell based on the reaction above?
- (A) Doubling the  $[Ag^+]$  from 1M to 2M  
 (B) Doubling the amount of M(s)  
 (C) Doubling the volume of the 1M  $Ag^+$  solution  
 (D) Reducing the  $[M^{2+}]$  from 1M to 0.5M
113. If a voltaic cell has a positive  $E^\circ$  value, what can be concluded about the values of  $\Delta G^\circ$  and  $K_{eq}$ ?
- (A)  $\Delta G^\circ < 0, K_{eq} < 1$               (B)  $\Delta G^\circ < 0, K_{eq} > 1$   
 (C)  $\Delta G^\circ > 0, K_{eq} < 1$               (D)  $\Delta G^\circ > 0, K_{eq} > 1$

114. A 3.00 amp current is used to electrolyze the molten chlorides;  $CaCl_2$ ,  $MgCl_2$ ,  $AlCl_3$ , and  $FeCl_3$ . The deposition of which mass of metal will require the longest electrolysis time?
- (A) 100 g Ca                      (B) 50 g Mg  
 (C) 75 g Al                        (D) 125 g Fe

115. When the reaction:  $Cl^- + ClO_3^- \rightarrow Cl_2 + H_2O$  is balanced in acid solution what is the ratio of  $Cl^-$  to  $ClO_3^-$ ?
- (A) 1 / 1              (B) 2 / 1              (C) 3 / 1              (D) 5 / 1

116. Which change could occur at the anode of an electrochemical cell?
- (A)  $Cl^- \rightarrow Cl_2$                       (B)  $H_2O \rightarrow H_2$   
 (C)  $Na^+ \rightarrow Na$                       (D)  $O_2 \rightarrow H_2O$

117.  $E^\circ = 0.93 \text{ V}$  for the reaction:
- | Standard Reduction Potential / $E^\circ$                     |
|--------------------------------------------------------------|
| $Fe^{2+}(aq) + 2e^- \rightarrow Fe(s) \quad -0.41 \text{ V}$ |
- $Fe(s) + 2M^+(aq) \rightarrow Fe^{2+}(aq) + 2M(s)$ .  
 What is the standard potential for  $M^+ + e^- \rightarrow M$ ?
- (A) 0.26 V                      (B) 0.52 V  
 (C) 0.67 V                      (D) 1.34 V

118. For which half-reaction will a 1.0 unit increase in pH cause the greatest increase in half-cell potential?
- (A)  $V^{2+}(aq) \rightarrow V^{3+}(aq) + e^-$   
 (B)  $VO_3^- + 2H^+ \rightarrow VO_2^+ + H_2O$   
 (C)  $VO^{2+} + 2H^+ + e^- \rightarrow V^{3+} + H_2O$   
 (D)  $VO^{2+} + H_2O \rightarrow VO_2^+ + 2H^+ + e^-$

119. A solution of aqueous  $CuSO_4$  is electrolyzed with a 1.50 ampere current for 30.0 minutes. What mass of copper metal is deposited?
- (A) 0.889 g              (B) 1.19 g              (C) 1.78 g              (D) 3.56 g

120. According to the tabulated standard reduction potentials

|                                                  |                             |
|--------------------------------------------------|-----------------------------|
| $Mg^{2+}(aq) + 2e^- \rightarrow Mg(s)$           | $E^\circ = -2.38 \text{ V}$ |
| $2H_2O(l) + 2e^- \rightarrow H_2(g) + 2OH^-(aq)$ | $E^\circ = -0.83 \text{ V}$ |
| $Br_2(l) + 2e^- \rightarrow 2Br^-(aq)$           | $E^\circ = 0.53 \text{ V}$  |
| $O_2(g) + 4H^+(aq) \rightarrow 2H_2O(l)$         | $E^\circ = 1.23 \text{ V}$  |

what products are formed during the electrolysis of an aqueous  $MgBr_2$  solution?

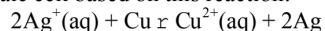
- (A) Mg and  $H_2$                       (B)  $H_2$  and  $Br_2$   
 (C)  $H_2$  and  $O_2$                       (D) Mg and  $O_2$

121. The equation for one of the half-reactions in a lead storage battery is:
- $$PbO_2 + 4H^+ + SO_4^{2-} + 2e^- \rightarrow PbSO_4 + 2H_2O$$
- What happens to the properties of the electrolyte as this cell discharges?

- |     |           |           |
|-----|-----------|-----------|
|     | Density   | pH        |
| (A) | increases | increases |
| (B) | increases | decreases |
| (C) | decreases | decreases |
| (D) | decreases | increases |

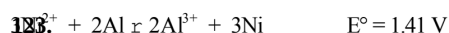
# Oxidation - Reduction

122. For the voltaic cell based on this reaction:



the concentrations of the aqueous ions and sizes of the electrodes can be changed independently. Which statement is correct?

- (A) Increasing the  $[\text{Cu}^{2+}]$  two-fold has the same effect on the cell voltage as increasing the  $[\text{Ag}^+]$  four-fold.
- (B) Decreasing the  $[\text{Cu}^{2+}]$  ten-fold has the same effect on the cell voltage as decreasing the  $[\text{Ag}^+]$  by the same ratio.
- (C) Decreasing the  $[\text{Cu}^{2+}]$  ten-fold has less effect on the cell voltage than decreasing the  $[\text{Ag}^+]$  by the same amount.
- (D) Doubling the sizes of the cathode has exactly the same effect on the cell voltage as decreasing the  $[\text{Cu}^{2+}]$  by a factor of two.



For the reaction given, which expression gives the value of  $\Delta G^\circ$  in  $\text{kJ}\cdot\text{mol}^{-1}$ ?

- (A)  $\frac{-3 \times 96.5}{1.41}$
- (B)  $\frac{-6 \times 96.5}{1.41}$
- (C)  $-3 \times 96.5 \times 1.41$
- (D)  $-6 \times 96.5 \times 1.41$

124. In which species is the oxidation number for hydrogen different from those in the other three?

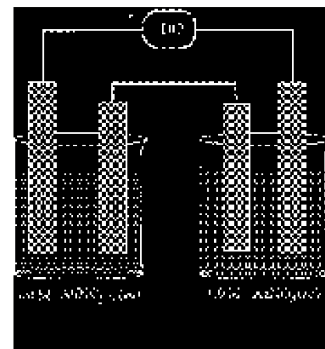
- (A)  $\text{AlH}_3$
- (B)  $\text{H}_3\text{AsO}_4$
- (C)  $\text{H}_3\text{PO}_3$
- (D)  $\text{NH}_3$

125. A solution containing equimolar amounts of  $\text{NiCl}_2$  and  $\text{SnBr}_2$  is electrolyzed using a 9V battery and graphite electrodes. What are the first products formed?

| Standard Reduction Potential (V)                                                  |        |
|-----------------------------------------------------------------------------------|--------|
| $\text{Ni}^{2+}(\text{aq}) + 2\text{e}^- \rightleftharpoons \text{Ni}(\text{s})$  | -0.236 |
| $\text{Sn}^{2+}(\text{aq}) + 2\text{e}^- \rightleftharpoons \text{Sn}(\text{s})$  | -0.141 |
| $\text{Br}_2(\text{aq}) + 2\text{e}^- \rightleftharpoons 2\text{Br}^-(\text{aq})$ | 1.077  |
| $\text{Cl}_2(\text{aq}) + 2\text{e}^- \rightleftharpoons 2\text{Cl}^-(\text{aq})$ | 1.360  |

- (A)  $\text{Ni}(\text{s})$  at cathode,  $\text{Cl}_2(\text{aq})$  at anode
- (B)  $\text{Ni}(\text{s})$  at cathode,  $\text{Br}_2(\text{aq})$  at anode
- (C)  $\text{Sn}(\text{s})$  at cathode,  $\text{Br}_2(\text{aq})$  at anode
- (D)  $\text{Sn}(\text{s})$  at cathode,  $\text{Cl}_2(\text{aq})$  at anode

126. True statements about the system shown after the passage of one Faraday of electricity include which of those given?



- I. The number of moles of Al formed is greater than the number of moles of silver formed.
- II. The final  $[\text{Al}^{3+}]$  is greater than the final  $[\text{Ag}^+]$ .
- III. The number of electrons reacting with  $\text{Al}^{3+}$  ions is the same as the number reacting with  $\text{Ag}^+$  ions.

- (A) I only
- (B) I and III only
- (C) II and III only
- (D) I, II and III

127. Which oxide is the best reducing agent?

- (A)  $\text{CO}_2$
- (B)  $\text{NO}_2$
- (C)  $\text{SiO}_2$
- (D)  $\text{SO}_2$

128. Which statement is correct about the electrochemical cell represented here?  $\text{Ag} | \text{Ag}^+ || \text{NO}_3^-, \text{NO} | \text{Pt}$

- (A) NO undergoes oxidation at the anode.
- (B) The major purpose of the Pt is to act as a catalyst.
- (C) The Ag electrode decreases in mass as the cell operates.
- (D) The voltage of the cell can be increased by doubling the size of the Ag electrode.

129. The overall reaction for the lead storage battery when it discharges is:  

$$\text{Pb}(\text{s}) + \text{PbO}_2(\text{s}) + 4\text{H}^+(\text{aq}) + 2\text{SO}_4^{2-}(\text{aq}) \rightarrow 2\text{PbSO}_4(\text{s}) + 2\text{H}_2\text{O}(\text{l})$$

- I.  $\text{PbSO}_4$  is formed only at the cathode.
- II. The density of the solution decreases.

Which statement(s) correctly describe(s) the battery as it discharges?

- (A) I only
- (B) II only
- (C) both I and II
- (D) neither I nor II

130. The standard reduction potential for  $\text{H}^+(\text{aq})$  is 0.00 V. What is the reduction potential for a  $1 \times 10^{-3} \text{ M HCl}$  solution?

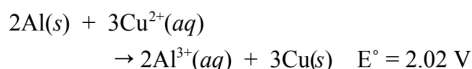
- (A) 0.355 V
- (B) 0.178 V
- (C) -0.178 V
- (D) -0.355 V

# Oxidation - Reduction

131. What is the approximate value of the equilibrium constant,  $K_{eq}$ , at 25 °C for the reaction;  
 $3Ag^+(aq) + Cr(s) \rightarrow Cr^{3+}(aq) + 3Ag(s)$   
 (A)  $10^{22}$  (B)  $10^{26}$  (C)  $10^{33}$  (D)  $10^{78}$
- | Standard Reduction Potential, V        |       |
|----------------------------------------|-------|
| $Ag^+(aq) + e^- \rightarrow Ag(s)$     | +0.80 |
| $Cr^{3+}(aq) + 3e^- \rightarrow Cr(s)$ | -0.74 |
132. Which products are formed by the electrolysis of an aqueous solution of  $AlCl_3$ ?  
 I.  $Al(s)$  II.  $Cl_2(g)$  III.  $H_2(g)$  IV.  $O_2(g)$   
 (A) I and III only (B) I and IV only  
 (C) II and III only (D) II and IV only
133. A current of 0.20 amps is passed through an aqueous solution of nickel(II) nitrate for 45.0 minutes. What mass of Ni metal (in grams) will be deposited?  
 (A) 0.16 (B) 0.22 (C) 0.33 (D) 0.66
134. What is the coefficient for  $OH^-$  after the equation  
 $\_ Br_2 + \_ OH^- \rightarrow \_ Br^- + \_ BrO_3^- + \_ H_2O$   
 is balanced with the smallest integer coefficients?  
 (A) 3 (B) 6 (C) 12 (D) 18
135. Which set of relationships could apply to the same electrochemical cell?  
 (A)  $\Delta G^\circ > 0$ ;  $E^\circ = 0$  (B)  $\Delta G^\circ < 0$ ;  $E^\circ = 0$   
 (C)  $\Delta G^\circ > 0$ ;  $E^\circ > 0$  (D)  $\Delta G^\circ < 0$ ;  $E^\circ > 0$
136. What is the standard cell potential for the reaction,  
 $2Cr(s) + 3Sn^{2+}(aq) \rightarrow 3Sn(s) + 2Cr^{3+}(aq)$   
 given the  $E^\circ$  values shown?  
 $Cr^{3+}(aq) + 3e^- \rightarrow Cr(s)$  -0.744 V  
 $Sn^{2+}(aq) + 2e^- \rightarrow Sn(s)$  -0.141 V  
 (A) 0.945 V (B) 0.603 V  
 (C) -0.603 V (D) -0.945 V
137. How many electrons are needed in the balanced half-reaction for the oxidation of ethanol to acetic acid?  
 $C_2H_5OH \rightarrow CH_3COOH$   
 (A) 1 (B) 2 (C) 3 (D) 4
138. Which is the weakest oxidizing agent in a 1 M aqueous solution?  
 (A)  $Ag^+(aq)$  (B)  $Cu^{2+}(aq)$   
 (C)  $H^+(aq)$  (D)  $Zn^{2+}(aq)$
139. The standard potential for the reaction  
 $Cl_2(g) + 2Br^-(aq) \rightarrow Br_2(l) + 2Cl^-(aq)$   
 is 0.283 volts. What is the equilibrium constant for this reaction at 25 °C?  
 (A)  $1.6 \times 10^{-5}$  (B) 22  
 (C)  $6.1 \times 10^4$  (D)  $3.8 \times 10^9$
140. When an aqueous solution of potassium fluoride is electrolyzed, which of the following occurs?  
 (A)  $O_2$  and  $H^+$  are produced at one electrode and  $H_2$  and  $OH^-$  are formed at the other.  
 (B)  $O_2$  and  $OH^-$  are produced at one electrode and  $H_2$  and  $H^+$  are formed at the other.  
 (C) Metallic K is formed at one electrode and  $O_2$  and  $H^+$  are formed at the other.  
 (D) Metallic K is produced at one electrode and elemental  $F_2$  is produced at the other.
141. A  $CuSO_4$  solution is electrolyzed for 20. minutes with a current of 2.0 ampere. What is the maximum mass of copper that could be deposited?  
 (A) 0.20 g (B) 0.40 g (C) 0.79 g (D) 1.6 g
142. Which anion can undergo both oxidation and reduction?  
 (A)  $Cr_2O_7^{2-}$  (B)  $NO_3^-$   
 (C)  $Cl^-$  (D)  $S^{2-}$
143.  $Cl_2 + OH^- \rightarrow Cl^- + ClO_3^-$   
 What is the coefficient for  $OH^-$  when this equation is balanced with the smallest integer coefficients?  
 (A) 2 (B) 3 (C) 4 (D) 6
144. Use the standard reduction potentials;  
 $Sn^{2+}(aq) + 2e^- \rightarrow Sn(s)$   $E^\circ = -0.141$  V  
 $Ag^+(aq) + e^- \rightarrow Ag(s)$   $E^\circ = 0.800$  V  
 to calculate  $E^\circ$  for the reaction;  
 $Sn(s) + 2Ag^+(aq) \rightarrow Sn^{2+}(aq) + 2Ag(s)$   
 (A) 0.659 V (B) 0.941 V  
 (C) 1.459 V (D) 1.741 V
145. Which of the processes happen during the discharging of a lead storage battery?  
 1.  $H_2(g)$  is produced  
 2.  $PbO_2$  is converted to  $PbSO_4$   
 3. The density of the electrolyte solution decreases  
 (A) 1 only (B) 2 only  
 (C) 1 and 3 only (D) 2 and 3 only

# Oxidation - Reduction

146. What is the value of  $\Delta G^\circ$  for the reaction?



- (A) -1170 kJ (B) -585 kJ  
(C) -390 kJ (D) -195 kJ

147. The voltage for the cell

$\text{Fe} | \text{Fe}^{2+}(0.0010 \text{ M}) || \text{Cu}^{2+}(0.10 \text{ M}) | \text{Cu}$   
is 0.807 V at 25 °C. What is the value of  $E^\circ$ ?

- (A) 0.629 V (B) 0.689 V  
(C) 0.748 V (D) 0.866 V

148. A current of 2.0 A is used to plate Ni(s) from 500 mL of a 1.0 M  $\text{Ni}^{2+}(aq)$  solution. What is the  $[\text{Ni}^{2+}]$  after 3.0 hours?

- (A) 0.39 M (B) 0.46 M (C) 0.78 M (D) 0.89 M

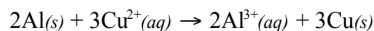
149. What is the average oxidation number of tungsten in the ion,  $\text{W}_6\text{O}_6\text{Cl}_{12}^{2-}$ ?

- (A) 2.7 (B) 3.3 (C) 3.7 (D) 4.3

150. How many moles of electrons must be removed from each mole of toluene,  $\text{C}_6\text{H}_5\text{CH}_3$ , when it is oxidized to benzoic acid,  $\text{C}_6\text{H}_5\text{COOH}$ ?

- (A) 1 (B) 2 (C) 4 (D) 6

Questions 151 and 152 refer to the reaction represented by this equation.



151. What is the value of  $E^\circ$  for a voltaic cell based on this reaction?

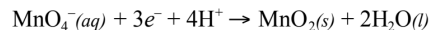
| Reaction                                             | $E^\circ$ |
|------------------------------------------------------|-----------|
| $\text{Cu}^{2+}(aq) + 2e^- \rightarrow \text{Cu}(s)$ | +0.34 V   |
| $\text{Al}^{3+}(aq) + 3e^- \rightarrow \text{Al}(s)$ | -1.66 V   |

- (A) 1.32 V (B) 2.00 V  
(C) 2.30 V (D) 4.34 V

152. What value should be used for  $n$  in the Nernst equation to determine the effect of changes in  $\text{Al}^{3+}(aq)$  and  $\text{Cu}^{2+}(aq)$  concentrations in this reaction?

- (A) 6 (B) 5 (C) 3 (D) 2

153. Use the given standard reduction potentials to determine the reduction potential for this half-reaction.



| Reaction                                                                                          | $E^\circ$ |
|---------------------------------------------------------------------------------------------------|-----------|
| $\text{MnO}_4^-(aq) + e^- \rightarrow \text{MnO}_4^{2-}(aq)$                                      | +0.564 V  |
| $\text{MnO}_4^{2-}(aq) + 2e^- + 4\text{H}^+ \rightarrow \text{MnO}_2(s) + 2\text{H}_2\text{O}(l)$ | +2.261 V  |

- (A) 1.695 V (B) 2.825 V  
(C) 3.389 V (D) 5.086 V

154. How many Faradays are required to reduce all the chromium in 0.150 L of 0.115 M of  $\text{Cr}_2\text{O}_7^{2-}$  to  $\text{Cr}^{2+}$ ?

- (A) 0.920 F (B) 0.690 F  
(C) 0.138 F (D) 0.069 F

155.  $\text{P}_4(s) + 3\text{OH}^-(aq) + 3\text{H}_2\text{O}(l) \rightarrow \text{PH}_3(g) + 3\text{H}_2\text{PO}_2^-(aq)$   
For this reaction, the oxidizing and reducing agents are, respectively,

- (A)  $\text{P}_4$  and  $\text{OH}^-$  (B)  $\text{OH}^-$  and  $\text{P}_4$   
(C)  $\text{P}_4$  and  $\text{H}_2\text{O}$  (D)  $\text{P}_4$  and  $\text{P}_4$

156.  $\text{Al}^{3+}(aq) + 3e^- \rightarrow \text{Al}(s) \quad E^\circ = -1.66 \text{ V}$   
 $\text{Cu}^{2+}(aq) + 2e^- \rightarrow \text{Cu}(s) \quad E^\circ = +0.34 \text{ V}$

What voltage is produced under standard conditions by combining the half-reactions with these Standard Electrode Potentials?

- (A) 1.32 V (B) 2.00 V (C) 2.30 V (D) 4.34 V

157. For which of these oxidation/reduction pairs will the reduction potential vary with pH?

I.  $\text{AmO}_2^{2+}/\text{AmO}_2^+$  II.  $\text{AmO}_2^{2+}/\text{Am}^{4+}$  III.  $\text{Am}^{4+}/\text{Am}^{2+}$

- (A) I only (B) II only  
(C) I and II only (D) I, II, and III

158.  $2\text{Ag}^+(aq) + \text{Cu}(s) \rightarrow \text{Cu}^{2+}(aq) + 2\text{Ag}(s)$   
The standard potential for this reaction is 0.46 V. Which change will increase the potential the most?

- (A) doubling the  $[\text{Ag}^+]$   
(B) halving the  $[\text{Cu}^{2+}]$   
(C) doubling the size of the  $\text{Cu}(s)$  electrode  
(D) decreasing the size of the  $\text{Ag}$  electrode by one half

159.  $10\text{Cl}^-(aq) + 2\text{MnO}_4^-(aq) + 16\text{H}^+(aq) \rightarrow 5\text{Cl}_2(g) + 2\text{Mn}^{2+}(aq) + 8\text{H}_2\text{O}(l)$

The value of  $E^\circ$  for this reaction at 25 °C is 0.15 V. What is the value of  $K$  for this reaction?

- (A)  $2.4 \times 10^{25}$  (B)  $4.9 \times 10^{12}$   
(C)  $1.2 \times 10^5$  (D)  $3.4 \times 10^2$

# Oxidation - Reduction

160. When water is electrolyzed, hydrogen and oxygen gas are produced. If 1.008 g of  $H_2$  is liberated at the cathode, what mass of  $O_2$  is formed at the anode?

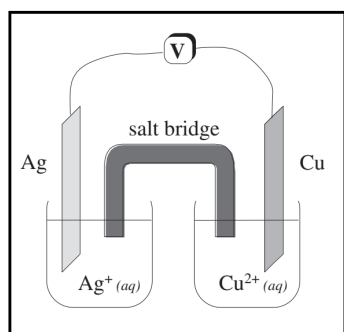
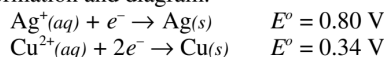
(A) 32.0 g (B) 16.0 g  
(C) 8.00 g (D) 4.00 g

161. What takes place when zinc metal is added to a aqueous solution containing magnesium nitrate and silver nitrate?

1. Zn is oxidized.
2.  $Mg^{2+}$  is reduced.
3.  $Ag^+$  is reduced.
4. No reaction takes place.

(A) 1 and 2 only (B) 1 and 3 only  
(C) 1, 2, and 3 only (D) 4 only

Questions 162, 163, and 164 should be answered with reference to this information and diagram.



162. What is the value for  $\Delta G^\circ$  when  $[Ag^+] = [Cu^{2+}] = 1.0 \text{ M}$ ?

(A) -44.4 kJ (B) -88.8 kJ  
(C) -243 kJ (D) -374 kJ

163. Which expression gives the voltage for this cell if  $[Cu^{2+}] = 1.00 \text{ M}$  and  $[Ag^+] = 0.010 \text{ M}$ ?

(A)  $0.46 \text{ V} + 0.0591 \text{ V}$   
(B)  $0.46 \text{ V} + 2 \times 0.0591 \text{ V}$   
(C)  $0.46 \text{ V} - 0.0591 \text{ V}$   
(D)  $0.46 \text{ V} - 2 \times 0.0591 \text{ V}$

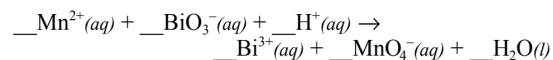
164. Which increases immediately if the surface area of the silver electrode is increased?

(A) overall cell voltage  
(B) rate of change of  $[Ag^+]$   
(C) mass of Cu electrode  
(D) change in ratio of electrode masses;  $\Delta \left( \frac{\text{mass of Cu}}{\text{mass of Ag}} \right)$

165. In the galvanizing process, iron is coated with zinc. The resulting chemical protection is most similar to that provided when

(A) a magnesium bar is connected to an iron pipe.  
(B) an iron can is plated with tin.  
(C) copper pipes are connected using lead solder.  
(D) a copper pipe is covered with epoxy paint.

166. What is the coefficient for  $H^+(aq)$  when the equation is balanced with whole number coefficients?

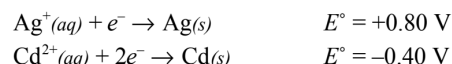


(A) 3 (B) 4 (C) 7 (D) 14

167. What is the sign of  $\Delta G^\circ$  and the value of  $K$  for an electrochemical cell for which  $E^\circ_{cell} = 0.80 \text{ V}$ ?

|     | $\Delta G^\circ$ | $K$   |
|-----|------------------|-------|
| (A) | —                | $> 1$ |
| (B) | +                | $> 1$ |
| (C) | +                | $< 1$ |
| (D) | —                | $< 1$ |

168. Consider a voltaic cell based on these half-cells.



Identify the anode and give the voltage of this cell under standard conditions.

(A) Ag;  $E_{cell} = 0.40 \text{ V}$  (B) Ag;  $E_{cell} = 2.00 \text{ V}$   
(C) Cd;  $E_{cell} = 1.20 \text{ V}$  (D) Cd;  $E_{cell} = 2.00 \text{ V}$

169. Which two species react spontaneously?

(A)  $Cu(s) + Ag^+(aq)$  (B)  $Br_2(l) + Cl^-(aq)$   
(C)  $H_2O(l) + Ca^{2+}(aq)$  (D)  $Au(s) + Mg^{2+}(aq)$

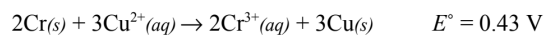
170. When aluminum oxide is electrolyzed in the industrial process for the production of aluminum metal, aluminum is produced at one electrode and oxygen gas is produced at the other. For a given quantity of electricity, what is the ratio of moles of aluminum to moles of oxygen gas?

(A) 1:1 (B) 2:1 (C) 2:3 (D) 4:3

# Oxidation - Reduction

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Questions 171. and 172. should be answered with reference to the reaction.



171. Which expression gives the value for  $\Delta G^\circ$  in  $\text{kJ}\cdot\text{mol}^{-1}$  for this reaction at  $25^\circ\text{C}$ ?

(A)  $-6 \times 8.31 \times 0.43 \times 1000$

(B)  $\frac{-6 \times 96500 \times 0.43 \times 1000}{8.31}$

(C)  $\frac{-6 \times 96500 \times 0.43}{1000}$

(D)  $\frac{-6 \times 8.31 \times 0.43}{1000}$

172. What is the voltage for this cell when  $[\text{Cu}^{2+}] = 1.0 \text{ M}$  and  $[\text{Cr}^{3+}] = 0.010 \text{ M}$ ?

(A) 1.2      (B) 0.87      (C) 0.47      (D) 0.39